1. Find the values of $a$ and $b$ if the function $f(x)=\begin{cases} -x^2 & x<0 \\ ax+b & 0\le x<1 \\ \sqrt{x+3} & x\ge 1 \end{cases}$ is continuous.

2. Let $f(x)=\dfrac{x^2-9}{x}$. State the discontinuities of $f(x)$ and find their type.

3. A designer is experimenting with a cylindrical can with a fixed height of 15 cm. find the rate of change of volume with respect to radius when the radius is 4 cm. the volume of a cylinder is $V=\pi r^2 h$ .

4. Let $f(x)=ax^2+bx+c$ . Find a,b and c so that the tangent to the graph $y=f(x)$ has slope 16 where $x=2$ and has x-intercepts (0,0) and (8,0)

5. Is the following statement true? Justify your answer.

***It is possible for a limit to exist even though the function may not be continuous at the point of question.***

6. Function $f(x)=\begin{cases} \dfrac{x^2-1}{x+1} & if \quad x\ne -1 \\ -2k+1 & if \quad x=-1 \end{cases}$ is continuous at $x=-1$.Find the value of k.

7. At what point on the parabola $y=3x^2$ is the slope of the tangent equal to 24?

8. Find the points on the curve $y=1-\dfrac{1}{x}$ where the tangent line is perpendicular to the line $y=1-4x$.

9. Find the average rate of change of the function $f(x)=\dfrac{5\sqrt{x}}{x+2}$ between $x=1$ and $x=4$ .What is the instantaneous rate of change at x= 1.

10. Evaluate the following limits, if they exist. Show your work.

a) $\lim\limits_{x\to\frac{2}{3}y}\dfrac{27x^3-8y^3}{9x^2-4y^2}$ b) $\lim\limits_{x\to\infty}7^{-(x-3)}$ c) $\lim\limits_{x\to 0}5^{x-3}$ d) $\lim\limits_{x\to 2}\dfrac{x^3-x^2-4x+4}{x^2-4}$

e) $\lim\limits_{x\to 0}\dfrac{\sqrt{4+x+x^2}-2}{x}$ f) $\lim\limits_{x\to\infty}\dfrac{\left(\frac{2}{3}\right)^x-2}{\left(\frac{2}{3}\right)^x+2}$ g) $\lim\limits_{x\to 4^+}\dfrac{x-4}{\sqrt{x}-2}$ h) $\lim\limits_{x\to 2}\dfrac{\sqrt{x^2+5}-\sqrt{x+7}}{x-2}$

i) $\lim\limits_{x\to 2}\dfrac{\sqrt[3]{2x-5}+1}{x-2}$ j) $\lim\limits_{x\to\infty}\dfrac{(2x+3)^2}{5-2x-5x^2}$ k) $\lim\limits_{x\to 2}\dfrac{\frac{1}{x-5}+\frac{1}{3}}{x-2}$

11. The position of a particle moving along the x-axis is given by s(t) = $t^2 - 5t + 4$ where t is the elapsed time in seconds.

(a) What is the position of the particle after 2 seconds?

(b) Calculate the average velocity of the particle from t = 2s to t = 5s.

(c) Calculate the instantaneous velocity of the particle when t = 2s.

12. Larry is driving over the speed limit on wet, slippery roads. The road can be modelled by the curve f(x) = $x^2$. As he travels from left to right, his car begins to slide out of the curve in a straight line at the point (-1, 1). A cat is sitting at the point (3, -7). Will Larry run over the cat?

13. Find the point(s) in exact value where the functions $y = 2x^3 + 10x - 1$ and $y = -\frac{4}{x}$ contain the same slope.

14. Determine the values of $b$ and $c$ in the function $f(x) = x^2 - 3bx + (c+2)$, if $f(x)$ has an x-intercept at $x = 1$ and a $f'(3) = 0$.

15. Use the given graph of $f(x)$ to graph $f'(x)$ on the same grid